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The effect of light on seed germination of *E. alba* Hassk

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Abstract-Controlled environmental chamber experiments were conducted for effect of light on seed germination on *E. alba* Hassk. and various experiments were set. The collected seeds were soaked overnight. The seed were placed on the seed container petriplates were kept in culture room. The observation was recorded to study germination and seedling emergence of *E. alba*. The plant *E. alba* is an important medicinal plant belonging to family Compositae and is used for producing Bhringraj Hair Oil. The seeds cultured on MS plain medium did not germinate. The seeds kept under dark condition did not germinate. The far-red light almost inhibits the process of Seed germination in *E. alba*. The rate of Seed germination varied from 0% - 6.7% under far-red light condition. The rate of Seed germination under normal light (white fluorescent light) was recorded to range from 65.9% to 76%. The Seed grown under red light showed 94% to 100% germination. Interestingly the Far Red treated seeds were found to germinate under Red light and the rate of germination was almost comparable to that of Red light treated seeds. The effects of red and far-red light are mutually reversible. At metabolic level, induction of germination should also initiate changes in the cell cycle, transcription and protein synthesis in the cells of embryos. The role of light was first of all studied in Grand Rapid cultivar of lettuce seeds. Plants have ability to change its metabolism, growth and development. The most important aspect of such adaptations results in initiating cell division from almost any tissue of the plant and to regenerate lost organs or undergo different developmental pathway in response to particular stimuli. Germination of seeds of a verity of plants is regulated by light is positively photoblastic seeds.

Keywords: Bhringraj, Far-red, Germination, transcription, Compositae

INTRODUCTION

Eclipta alba Hassk is an annual, dicot, herbaceous plant commonly known as King of hairs. *E. alba* is an important medicinal plant belonging to family Compositae (Asteraceae). And is used for producing Bhringraj Hair Oil. It is also known as *Eclipta erecta* or *Eclipta prostrata, Verbesina alba* or *Verbisina prostrata.* In other words, it is commonly known as False daisy, Yerba de tago bhringraj, Keshraj in Assamese and Karisalan Kanni in Tamil.

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The plant is erect or prostrate, much branched, roughly hairy, annual rooting at the nodes. The leaves are opposite, sessile and lanceolate, The leaves possess white appressed hair. It grows commonly in moist places as a weed all over the World. It is widely distributed throughout India, China, Thailand and Brazil. In India it is considered as a nuisance in rice crop. In other countries it is known to cause problems in other crops like Soybean and peanut.

The plant remains a major source of medicinally active principles.[1] The significance of *E. alba* as a medicinal plant has been widely mentioned in both ancient and modern literature.[1,4]

The important chemical constituent of *E. alba* are coumestans alkaloids, Flavonoids, Glycosides Poly acetylene. Plants have ability to change its metabolism, growth and development according to their environment. The germination of seeds has been the most fascinating and essential developmental episodes in the life cycle of flowering plant. Role of light, temp and hormones have been significant. Germination of seeds of a variety of plants is regulated by light is positively photoblastic seeds. *E. alba* seeds have an absolute requirement for light to germinate. Red light in particular, has been shown to promote germination and seedling growth in *E. alba*. Knowledge of the effect of light on germination is essential in the propagation of plant species for restoration purposes[5] and for a better understanding of germination ecology.[6]

The germination response to light may vary between habitats. Light is essential for photosynthesis and influences various physiological process such as stomatal function electrolyte absorption and Transportation[7] though this plant abundantly found as weed, some of the farmers have attempted to cultivate it for its medicinal properties. Its micro-propagation has also been attempted because of its demand in national and international market.[8] The main objective of the present investigation was to study the effect of light on seed germination of *E. alba* Hassk.

MATERIALS & METHODS

E. alba Hassk grows as a wild herb in the University campus. In the Flowering season the characteristic inflorescence (head) of appropriate age (immediately after anthesis) were collected and surface sterilized by immersion in 0.1% mercuric chloride for 5 min. followed by 4-5 washing in double distilled sterile water. The heads were cultured on liquid MS medium at different strength viz MS (full strength), MS 1/2, MS 1/4, using cotton as matrix. The cultures were maintained at 25°C ± 2°C Under light (64 μ E $m^{-2}s^{-1}$) in culture room.

The different spectral qualities of monochromatic blue, green, red and far red irradiance were produced by passing incandescent light through filter as per methods described by Mancenelli, *et al.*(1966)[9]. The blue, green, red and far red filters transmitted maximally at 438, 542, 664 and 735 nm respectively, but the actual energy at agar surface in Petridishes was 2.94×10^3 ergs cm^{-2} s^{-1} which is less than the energy available with the red light; none the less, this quantum of energy has been found by Haupt (1959)[10] to fall in the range Used in red: far-red reversible Studies in Eukaryotes. Experiments were also conducted to make reversal studies using red and far-red filter. The standard culture vessels was used of Pyrex or Borosil glass, The experiments were repeated five times to test the reproducibility of the results.

RESULT & DISCUSSION

The germination of seeds has been the most fascinating and essential developmental episodes in the life cycle of a flowering plant. The germination of seeds seems to be regulated by light temperature and hormones. At metabolic level, induction of germination should also initiate changes in the cell cycle, transcription and protein synthesis in the cells of embryos.[11] Germination of seeds of a variety of plants is regulated by light is positively photoblastic seeds.[12]

In order to study the effect of one of the most important environmental factors, light, on *E. alba*, various experiments were set. The collected seeds were soaked overnight. The seeds were placed on the seed container petriplates and kept in the culture room. Observations were recorded to study germination and seedling emergence of *E. alba* (Table 1-4). The seeds cultured on MS plain medium did not germinate. The seeds kept under dark conditions also did not germinate (Fig.1). The Far-Red almost inhibited the process of seed germination in *E. alba*. As shown in Table 2, the rate of seed germination varied from 0% - 6.7% under Far-Red light conditions. The rate of seed germination under normal light (white fluorescent light) was recorded to range from 65.9% to 76% (Fig. 2). The seeds grown under Red light showed 94% to 100% germination. Interestingly, the Far-Red treated seeds were found to germinate under Red light and the rate of germination was almost comparable to that of red light treated seeds. Similar observed when Far-Red light treated seeds were placed under red light germination of seeds were observed.

The study revealed the fact that a brief exposure of fully imbibed seeds to red light induces germination, while the red light followed by far-red light inhibits germination. The effects of red and far-red light are mutually reversible. The red-far-red effect is under regulation of a pigment called phytochrome. The phytochrome present in the dormant seed in the red absorbing form Pr when the seeds receive red light Pr is converted to Pfr. The dark germination of seeds may also be explained using this model due to rehydration of Pfr phytochrome present in them.

Table 1- Effect of Dark condition

Sl. No.	Number of seeds	Germination of seeds	Percentage of germination
1.	**62**	Not observed	No Germination
2.	**70**	Not observed	No Germination
3.	**42**	Not observed	No Germination
4.	**50**	Not observed	No Germination

Table 2- Effect of Far - Red light condition

Sl. No.	Number of seeds	Germination of seeds	Percentage of germination
1.	**41**	No seeds germinated	0%
2.	**47**	03 seeds germinated	6.30%
3.	**59**	04 seeds germinated	6.70%
4.	**47**	01 seeds germinated	2.120%
	Mean±S.D.=48.5±6.54		Mean±S.D.=3.78.5±2.82

Table 3- Effect of White light

Sl. No.	Number of seeds	Germination of seeds	Percentage of germination
1.	**44**	29 seeds germinated	65.90%
2.	**48**	36 seeds germinated	75.00%
3.	**53**	40 seeds germinated	75.40%
4.	**50**	38 seeds germinated	76.00%
	Mean±S.D.=48.75±3.27		Mean±S.D.=73.075±4.16

Table 4- Effect of Red light on seed germination

Sl. No.	Number of seeds	Germination of seeds	Percentage of germination
1.	**60**	58 seeds fully germinated	96.70%
2.	**50**	49 seeds fully germinated	98.00%
3.	**55**	Seeds completely germinated	100%
4.	**53**	50 seeds germinated	94.30%
	Mean±S.D.=54.5±3.64		Mean±S.D.=97.25±2.07





The phytochrome as a photoreceptor in positively photoblastic seeds is well established. An important questions needs to be answered as to how the Pfr form induces germination. According to Hendricks and Taylorson (1978)[13] the binding of pigment to the cell membrane might be the first event in the process leading to radicle elongation.

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